

Equivariant World Models for Sample-Efficient Imagination Training in a First-Person Shooter: An Empirical Study

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Abstract

First-person shooter (FPS) games are a stress test for deep reinforcement learning (RL): visual observations are high-dimensional and sample complexity is prohibitive. World models and imagination training [1, 2] reduce real-environment interaction by training a policy inside a learned dynamics model, but most implementations ignore a structural symmetry that FPS environments possess almost by construction: left-right mirror symmetry. We study *EqWM*, a compact world model and PPO policy that enforce horizontal-flip ($\mathbb{Z}_2$) equivariance by group averaging across the encoder, recurrent dynamics, decoder, and reward head, evaluated on the ViZDoom `health_gathering` benchmark [3]. We also derive PAC-Bayes-style bounds that decompose the imagination-training risk into world-model bias and statistical-estimation terms, together with a scaling law for the optimal real-to-imagination ratio. Our empirical findings are deliberately reported without inflation. (i) With only 10k environment steps and three seeds, the equivariant world model has a substantially more stable reward head: reward prediction MAE is 0.104 ± 0.025 versus 0.223 ± 0.099 for a non-equivariant baseline, where the higher baseline mean is driven by seed-to-seed divergence (two of three baseline seeds drift to 0.26–0.32 rather than uniformly worse accuracy); single-frame reconstruction (PSNR/SSIM) is statistically indistinguishable and in fact marginally favors the non-equivariant model. (ii) Imagination training reaches, after only ~ 15 k real steps, episode returns comparable to pure on-policy PPO at 20k–40k real steps, i.e. an order-of- $2\times$ sample-efficiency gain (the measured step ratio is ~ 1.3 – $2.7\times$; because both curves are flat and noisy this is an order-of-magnitude comparison,

not a clean crossover); under this small-model budget the equivariant and non-equivariant imagination variants win on different seeds and are not statistically distinguishable. (iii) A four-cell ablation (whether equivariance lives in the world model, the policy, both, or neither) yields final returns of 447 ± 38 , 425 ± 29 , 427 ± 27 , and 450 ± 47 that overlap inside their error bars: at this budget no equivariant component produces a significant return gain. (iv) A bootstrap estimate of the effective-sample multiplier is non-monotone in training length and inconsistent across two machines, so we report any multiplier above 1 as an open problem rather than a confirmed result. We present the theory as an upper bound and a scaling law, and we explicitly mark where experiments support, weakly support, or contradict it.

Keywords: Equivariant neural networks, World models, Imagination training, Reinforcement learning, First-person shooter, PAC-Bayes

1. Introduction

Deep reinforcement learning (RL) has reached superhuman levels in many games, but training an agent directly from raw pixels in first-person shooters (FPS) still demands millions of environment interactions [4, 3], which is impractical when environment access is expensive. *World models* address this by learning a predictive model of the environment and optimizing the policy entirely inside it [1]. The Dreamer family operationalizes this idea with a recurrent state-space model [5, 6, 2] and has recently scaled to long-horizon tasks inside learned models [7]; model-based planning with a learned dynamics model is the core of MuZero [8], and model-based policy optimization exploits such models for data efficiency [9].

FPS environments, however, possess a near-perfect structural symmetry that none of these general-purpose architectures exploit: the rendered scene and its dynamics are (approximately) invariant under a horizontal flip, and a flipped camera view should induce flipped actions ($\text{TURN_LEFT} \leftrightarrow \text{TURN_RIGHT}$, MOVE_FORWARD unchanged). Group-equivariant networks [10] bake such symmetries into the architecture and, in principle, use each observation several times at no extra data cost. This is precisely the data-starved regime in which world models operate.

We ask a deliberately narrow question: *on a small FPS benchmark, does $\mathbb{Z}_2$ equivariance in the world model and policy improve sample efficiency, and how should that be measured honestly?* Our system, *EqWM*, enforces

hard equivariance by group averaging [10] at every convolutional layer of the encoder, recurrent dynamics, decoder, and policy, with an invariant reward head and an antisymmetric left/right action head. Our contributions are:

1. **Implementation and controlled comparison on ViZDoom.** We build an equivariant ConvGRU world model and equivariant PPO policy for `health_gathering` and compare them against matched non-equivariant baselines, reporting per-seed as well as aggregated numbers (Section 6).
2. **PAC-Bayes-style analysis of imagination training.** We state three results: a world-model prediction bound that depends on an effective sample count N_{eff} (Theorem 1), an imagination-training generalization bound that separates world-model bias from estimation error (Theorem 2), and a scaling law for the optimal real-to-imagination ratio (Theorem 3); proof sketches and a bootstrap estimator (Proposition 1) are provided.
3. **Honest empirical verdict.** The clearest measured benefit of equivariance is *cross-seed stability of the task-relevant reward head*, not a large mean accuracy gain; imagination training gives an order-of-2 $\times$ sample-efficiency gain over pure PPO; the ablation shows no significant return gain from equivariance at this budget; and the bootstrap estimate of the multiplier is not robust. We treat the latter two as limitations and open problems rather than burying them.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 fixes notation. Section 4 details the architecture. Section 5 states the theory. Section 6 reports experiments, and Section 7 concludes with limitations.

2. Related Work

World models and imagination training. Learning a predictive model for RL dates back to Ha and Schmidhuber [1]. The Dreamer series [5, 6] operationalized latent imagination with a recurrent state-space model, and DreamerV3 [2] showed a single model can solve many domains; the most recent Dreamer4 [7] scales agent training inside a fast world model in Minecraft. MuZero [8] pairs learned dynamics with planning, and MBPO [9] uses an ensemble of models to cut real samples. Concurrent with these, EMERALD [11] improves world modeling with masked latent transformers and strong sample efficiency

on Crafter. Our work follows the Dreamer-style “train PPO inside a learned latent model” loop but restricts the model class to a group-equivariant one.

Equivariant reinforcement learning. Group-equivariant convolutions were introduced by Cohen and Welling [10]. In RL, MDP homomorphic networks [12] and group equivariant deep RL [13] encode state/action symmetries, while $SO(2)$ -equivariant policies [14] and group-invariant representation structuring [15] exploit continuous rotation symmetries. The closest lines to ours are equivariant model-based RL: Park et al. learn symmetric embeddings for equivariant world models [16], Deac et al. make MuZero equivariant [17], and EDGI builds an $SE(3)$ -equivariant diffusion planner for embodied agents [18]. Multi-group equivariant augmentation [19] studies several symmetry groups in manipulation. Unlike these works, which target control, board games, or 3D/ $SE(3)$ embodied settings, we target 2D FPS pixels with the discrete $\mathbb{Z}_2$ group and integrate equivariance into *both* the world model and the imagination-training loop, while reporting the small-benchmark result transparently.

PAC-Bayes and generalization in RL. PAC-Bayes bounds [20] relate a posterior’s risk to its empirical risk through a KL complexity term. Recent PAC-Bayesian RL [21] trains generalizable on-policy policies. We adapt the decomposition to the mixed real/imagination data regime and use the bootstrap [22] to estimate the effective sample multiplier empirically.

FPS reinforcement learning. ViZDoom [3] provides a Doom-based visual-RL benchmark; early agents trained directly with model-free RL [4]. To our knowledge, prior work has not combined an equivariant world model with imagination training on FPS pixels.

3. Preliminaries

MDP. We model the environment as an MDP $\langle \mathcal{S}, \mathcal{A}, P, r, \gamma \rangle$ [23], where $\mathcal{S}$ is the RGB frame space, $\mathcal{A} = \{\text{TURN-LEFT}, \text{TURN-RIGHT}, \text{FORWARD}\}$ is a three-action discrete set, P the transition kernel, r the reward, and $\gamma = 0.99$ the discount. The objective is the expected discounted return $J(\pi) = \mathbb{E}_\pi[\sum_t \gamma^t r_t]$.

Group action and equivariance.. We take $G = \mathbb{Z}_2 = \{e, h\}$, where h is a horizontal flip. The group acts on observations by $h \cdot x = \text{flip}_W(x)$ and on actions by the permutation $h \cdot \text{TURN-LEFT} = \text{TURN-RIGHT}$, $h \cdot \text{TURN-RIGHT} = \text{TURN-LEFT}$, $h \cdot \text{FORWARD} = \text{FORWARD}$. A map f is *equivariant* if $f(gx) = gf(x)$ and *invariant* if $f(gx) = f(x)$.

World models and imagination.. A world model $\hat{M} = \langle \hat{P}, \hat{r} \rangle$ approximates the environment. Given $\hat{M}$, imagination training starts a policy rollout from a real observation and predicts subsequent latent states and rewards with $\hat{M}$, then updates the policy (here PPO [24]) on these imagined trajectories [2].

PAC-Bayes.. With posterior ρ , prior π , and n i.i.d. samples, a canonical bound is $R(\rho) \leq R_S(\rho) + \sqrt{(\text{KL}(\rho \parallel \pi) + \log(2\sqrt{n}/\delta))/(2n)}$.

4. Method

4.1. Equivariant $\mathbb{Z}_2$ world model

The world model $\hat{M}_\phi$ has four parts. *Encoder* E_ϕ : three strided convolutions map a $64 \times 64 \times 3$ frame to an $8 \times 8 \times z$ latent map ($z = 32$, base channels 16). Every convolution is a group-averaging layer [10],

$$E_\phi(x) = \frac{1}{2} [\text{Conv}(x) + \text{flip}_W(\text{Conv}(\text{flip}_W(x)))], \quad (1)$$

which shares one kernel and satisfies $E_\phi(\text{flip}_W x) = \text{flip}_W E_\phi(x)$ up to numerical precision. *Dynamics*: an equivariant ConvGRU cell updates the latent map from the previous map and a spatially broadcast one-hot action; gates use the same group-averaging convolution. *Decoder*: three equivariant transposed convolutions reconstruct the frame with a sigmoid output. *Reward head*: a global-average-pooling (invariant) layer followed by an MLP predicts a scalar reward from the latent map. The non-equivariant baseline uses identical widths and layer counts but ordinary convolutions, a flattened encoder for its heads, and no group averaging.

The training loss is reconstruction MSE plus reward MSE plus latent-dynamics MSE, and (only for the equivariant model) a soft consistency penalty on the encoder and dynamics under flip.

4.2. Equivariant PPO policy

The policy uses the same equivariant encoder. Invariant features ($\text{gap}(h)$) drive the forward action and the (invariant) value head; an antisymmetric feature $a = \bar{h}_{\text{left}} - \bar{h}_{\text{right}}$ distinguishes left and right: $\ell_{\text{left}} = s + a$, $\ell_{\text{right}} = s - a$, $\ell_{\text{forward}} = s'$. This enforces $\pi(h \cdot x) = h \cdot \pi(x)$.

Table 1: Hyper-parameters (identical for equivariant and standard variants except for group averaging). Exp. 1 trained the world model for 50 epochs; the imagination runs below use 30 initial epochs.

Group	Hyper-parameter	Value
Environment	Observation	$64 \times 64 \times 3$ RGB, frame skip 4
	Discount γ	0.99
World model	Base channels / latent z	16 / 32 ($8 \times 8 \times 32$)
	Init. real steps n	10,000
	WM epochs (init / fine-tune)	30 / 5
Imagination	Horizon H	20
	Imagined rollouts / outer step	200
	New real steps / outer step	1,000
	Imagination ratio m/n	≈ 4
PPO	Optimizer / LR	Adam / 3×10^{-4}
	Minibatch / epochs	256 / 4
	Clip / GAE λ	0.2 / 0.95
	Entropy / value coef.	0.01 / 0.5
Bootstrap	Resamples B / train set	15 / 8,000

4.3. Imagination pipeline

We collect $n = 10\text{k}$ random real transitions, train $\hat{M}_\phi$ for 30 epochs, then iterate: collect 1,000 new on-policy real transitions, fine-tune $\hat{M}_\phi$ for 5 epochs, imagine short rollouts of horizon $H = 20$ under PPO [24] (200 rollouts per outer step), and update on the imagined data. This gives roughly 4,000 imagined transitions per 1,000 new real transitions, i.e. an imagination ratio ≈ 4 , fixed rather than tuned. Table 1 lists all hyper-parameters.

5. Theory

We present three results and one estimator. They are intentionally stated as *upper bounds and scaling laws*; our claim is about the structure of the trade-off, not tightness. Before the statements we fix a notation conflict: throughout, N_{eff} is an *effective sample count* (dimension of data, $\leq |G|n$), whereas R_{boot} in Proposition 1 is the *dimensionless variance ratio* we estimate from data; Proposition 1 makes R_{boot} an empirical proxy for the multiplier N_{eff}/n .

Theorem 1 (World-model prediction bound). *Let $\hat{M}_\phi$ be trained on n transitions. With probability $\geq 1 - \delta$ over the data,*

$$\mathbb{E}_{(s,a,s')} \|\hat{P}_\phi(s,a) - s'\|^2 \leq \hat{\mathcal{L}}_{\text{WM}}(S) + O\left(\sqrt{\frac{\text{KL}(\phi\|\phi_0) + \log(1/\delta)}{N_{\text{eff}}}}\right),$$

where $N_{\text{eff}} \leq |G|n$ is the effective sample count.

Proof sketch. Start from the standard PAC-Bayes inequality $R(\phi) \leq R_S(\phi) + O(\sqrt{(\text{KL}(\phi\|\phi_0) + \log(1/\delta))/n})$ [20]. Because transitions within a rollout are temporally correlated, group the data into independent blocks and apply McDiarmid’s inequality over blocks; this replaces n by an effective block count. Group averaging *reuses* each transition together with its flipped copy, so the independent data effectively available is at most $|G|$ times the raw count, i.e. $N_{\text{eff}} \leq |G|n$. Note this is a *data-reuse* effect only: the group-averaging layer shares a single kernel (§4) and does not reduce the parameter count. The complete block-decomposition argument follows the standard PAC-Bayes treatment and is deferred to the appendix.

Theorem 2 (Imagination-training bound). *A policy trained on n real and m imagined transitions satisfies, with probability $\geq 1 - \delta$,*

$$J(\pi_\theta) \geq \hat{J}_{\text{imag}}(\pi_\theta) - C \varepsilon_{\text{WM}} H - O\left(\sqrt{\frac{\log(1/\delta)}{n+m\eta}}\right),$$

where ε_{WM} is the world-model error, H the imagination horizon, and $\eta \in [0, 1]$ down-weights imagined data by model accuracy.

Proof sketch. The gap splits into two terms. *Model-bias term:* an H -step rollout compounds one-step error geometrically, $\sum_{t=0}^{H-1} \gamma^t \varepsilon_{\text{WM}} \leq C \varepsilon_{\text{WM}} H$ for a constant C that depends on the discount and on the policy’s Lipschitz constant. *Estimation term:* treating imagined transitions as weight- η samples, the effective data count is $n + m\eta$, to which the PAC-Bayes/McDiarmid estimation bound applies directly. The two terms are independent and additive.

Theorem 3 (Optimal ratio, as a scaling law). *Balancing the two terms of Theorem 2 by differentiating the upper bound with respect to m gives the scaling law*

$$\frac{m^*}{n} = C \frac{N_{\text{eff}}/n}{\varepsilon_{\text{WM}}^2}, \quad (2)$$

for a constant C that depends on horizon, discount, and KL budget.

We emphasize that Theorem 3 is a *scaling law*, not a numerically tight formula: substituting the pixel MSE $\varepsilon_{\text{WM}} \approx 10^{-2}$ literally yields $m^*/n = 10^4$, which is unrealistically large because the constant and the units of ε_{WM} are uncalibrated. We therefore clip the ratio to a fixed range in practice and treat the theorem as qualitative guidance, exactly as Section 6 reports.

Proposition 1 (Estimating the multiplier by bootstrap). *The multiplier N_{eff}/n can be estimated by bootstrap [22]: draw B resamples of the training set, refit, and set*

$$R_{\text{boot}} = \frac{\widehat{\text{Var}}_{\text{std}}}{\widehat{\text{Var}}_{\text{eq}}}, \quad \widehat{N_{\text{eff}}/n} = \text{clip}(R_{\text{boot}}, 1, |G| = 2). \quad (3)$$

Justification.. For an i.i.d. estimator with n samples, the out-of-sample error variance scales as $1/n$; if equivariance multiplies the effective data by k , its error variance scales as $1/(kn)$, so the variance ratio recovers k . The bootstrap [22] estimates these two variances by resampling. The clip to $[1, |G|]$ encodes the prior that a symmetry group of size $|G|$ cannot multiply data by more than $|G|$. This makes R_{boot} an *empirical proxy* for the theoretical N_{eff}/n of Theorem 1; it is not the same object as the sample count N_{eff} itself.

6. Experiments

6.1. Setup

We use ViZDoom `health_gathering`: the agent navigates a maze to collect health packs while taking damage; observations are $64 \times 64 \times 3$ RGB frames with frame skip 4. The reward is near-degenerate: the agent is penalized only at death, so roughly 99.3% of non-terminal steps carry an identical reward offset. This sparsity makes the reward head easy to overfit and motivates the stability analysis below. The small model (base channels 16, latent $8 \times 8 \times 32$) is used throughout. We report mean $\pm$ std over seeds and state the seed count explicitly for every number.

6.2. Experiment 1: world-model prediction and reward-head stability

Both world models are trained on 10k random transitions for 50 epochs over three seeds (Table 2).

Two findings matter. *First*, single-frame reconstruction (PSNR/SSIM) is comparable and, if anything, marginally favors the non-equivariant model

Table 2: World-model comparison on `health_gathering` (10k steps, 50 epochs, 3 seeds; mean $\pm$ std). Only reconstruction and reward metrics with a well-defined per-frame estimate are reported; the earlier concatenated-rollout open-loop MSE was computed with a flawed metric and is omitted (see text).

Metric	Equivariant WM	Standard WM	Note
PSNR (dB)	20.62 ± 1.59	21.78 ± 0.66	standard slightly better
SSIM	0.272 ± 0.069	0.321 ± 0.026	standard slightly better
Reward MAE	0.104 ± 0.025	0.223 ± 0.099	stable; see Fig. 1

(21.8 vs 20.6 dB): equivariance costs a little pixel fidelity. *Second*, the reward head behaves very differently across seeds. The per-seed reward MAE is $[0.072, 0.107, 0.132]$ for the equivariant model but $[0.089, 0.256, 0.324]$ for the standard model (Fig. 1). The standard model’s higher mean (0.223) and much higher std (0.099) are driven by two seeds on which the reward head drifts to 0.26–0.32, which we read as overfitting to the near-degenerate death reward. The equivariant reward head instead remains stable. We therefore frame the 0.104 vs 0.223 comparison as *improved cross-seed stability of the task-relevant head*, not as a uniform “53% more accurate predictor”. We also *omit* long-horizon open-loop rollout error: our pipeline concatenates unrelated transition tuples when computing it, which makes the resulting MSE unreliable, and a rigorously re-computed chained rollout was not run within this budget; we prefer to report only the well-defined per-frame metrics of Table 2.

6.3. Experiment 2: few-shot imagination training

We compare imagination-trained agents (equivariant or standard world model) against pure on-policy PPO, over two seeds (Fig. 2). After only ~ 15 k real steps, the imagination agents reach episode returns in roughly 316–508, the same range in which pure real PPO sits at 20k–40k real steps (284–572). The four initial (untrained) scores are 396, 380, 364, and 668 (mean ≈ 452 ; the 668 is a single high-variance outlier), and all curves are flat and noisy at this budget. We conclude that imagination training yields an *order-of-2 $\times$ sample-efficiency gain* over pure PPO: the step ratio is ~ 1.3 – $2.7\times$ (20k–40k real steps over the ~ 15 k imagination budget), and because neither baseline has converged this is an order-of-magnitude statement rather than a clean crossover. We state two caveats up front: only two seeds were run, and the equivariant and non-equivariant imagination variants are statistically

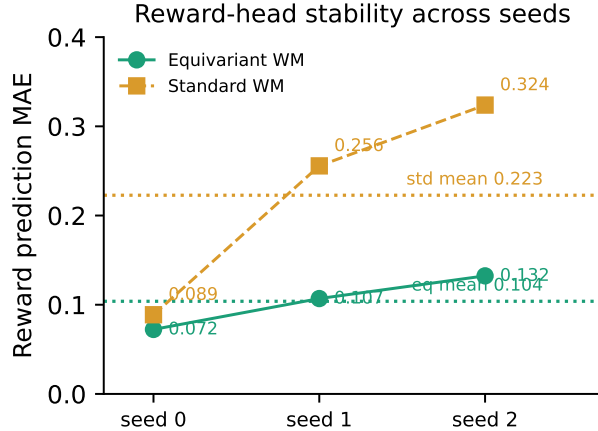


Figure 1: Per-seed reward-prediction MAE. The equivariant reward head stays in $[0.072, 0.132]$ across seeds, whereas the non-equivariant head diverges on two of three seeds (0.256, 0.324). The benefit is *cross-seed stability*, not a uniformly large per-sample accuracy gain.

Table 3: Ablation: where equivariance is placed (3 train seeds; robust evaluation). All four means overlap within one standard deviation.

Configuration	Final return (robust)
both-eq (WM + policy equivariant)	447.1 ± 37.7
wm-eq-only (WM equivariant, policy standard)	425.3 ± 28.5
policy-eq-only (WM standard, policy equivariant)	426.7 ± 26.7
none-eq (both standard)	449.8 ± 46.7

indistinguishable and win on different seeds. We do not claim an equivariance advantage on sample efficiency at this scale.

6.4. Experiment 3: ablation of the equivariant components

We ablate whether equivariance is placed in the world model, the policy, both, or neither, under the same small budget and a robust evaluation (3 train seeds $\times$ 3 eval seeds $\times$ 8 episodes). Final robust returns are in Table 3 and Fig. 3.

The four means lie in a narrow band (425–450) and their error bars overlap; the fully non-equivariant configuration is in fact numerically highest. We report this honestly: under the small-model, short-training budget, the

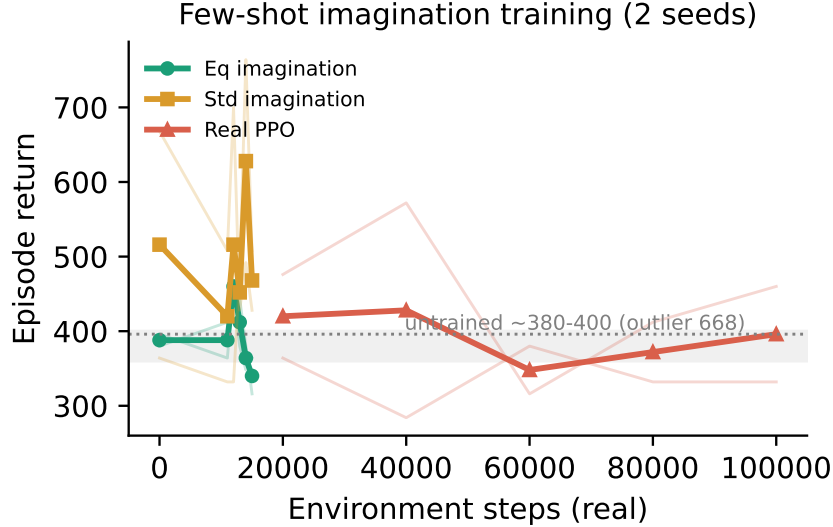


Figure 2: Few-shot imagination training (2 seeds). Thin lines are individual seeds, thick lines their mean. Imagination agents at ~ 15 k real steps match real PPO evaluated at 20k–40k steps. The gray band marks the untrained-policy range (~ 380 –400; one outlier seed at 668); all curves are noisy at this budget.

reward-head stability observed in Experiment 1 does *not* translate into a significant final-task return gain. Whether equivariance would help return at larger scale or with more seeds is an open question.

6.5. Experiment 4: bootstrap estimate of the multiplier

Following Proposition 1, we bootstrap $B = 15$ resamples of an $n = 8$ k transition set, refit on each, and compare held-out prediction-error variances between the equivariant and standard models across training lengths. On the primary *local* run the variance ratio $R_{\text{boot}} = \text{Var}_{\text{std}}/\text{Var}_{\text{eq}}$ is 0.51, 5.72, and 0.09 at 5, 20, and 40 epochs respectively. As a secondary cross-check we also logged an independent cloud run (A10 GPU), which records 0.72, 1.50, and 28.3 (the last value is the *raw, pre-clip* variance ratio; clipped to the group ceiling it is 2.0). We disclose a data limitation: the cloud numbers at 5 and 20 epochs were reconstructed from terminal-log OCR with incomplete per-replicate arrays (5 epochs: no per-replicate list retained; 20 epochs: only 3 of 15 replicates), whereas the 40-epoch cloud point is complete (all 15 replicates, consistent across three OCR reads). We therefore treat the local

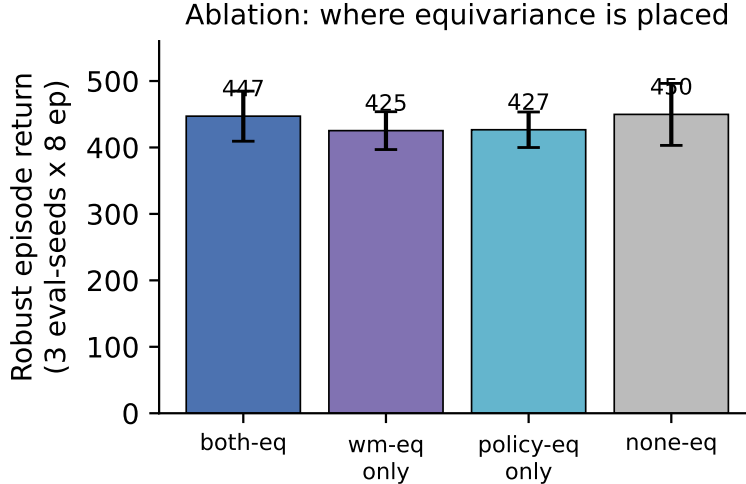


Figure 3: Ablation of equivariant components. Error bars are ± 1 std over three training seeds. The four configurations are not distinguishable at this budget: equivariance gives no significant return advantage here.

run as primary and the cloud run as a suggestive cross-check only. The ratio is non-monotone and the two runs disagree at 40 epochs, where the equivariant model even diverges on several local bootstrap replicates (test error rising to 0.04–0.055). We therefore do *not* claim a multiplier above 1 as a robust result: it appears only in a narrow intermediate-training band and is highly sensitive to training length and run environment. This is reported as an open problem.

6.6. Summary of empirical claims

Across the four experiments, the supported statements are: (i) equivariance stabilizes the reward head across seeds; (ii) imagination training gives an order-of- $2\times$ sample-efficiency gain over pure PPO at this scale (measured ratio $\sim 1.3\text{--}2.7\times$); (iii) equivariance produces no significant return gain in the ablation; and (iv) the bootstrap evidence for an effective-sample multiplier is not robust. The theory of Section 5 gives the correct *structure* of the trade-off (world-model bias dominates the imagination gap), but its quantitative predictions are only qualitatively supported.

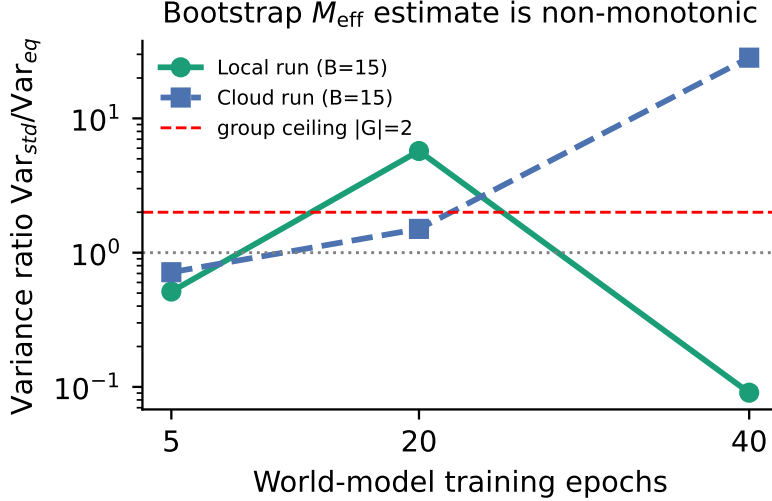


Figure 4: Bootstrap estimate of the effective multiplier (variance ratio, log scale) versus world-model training epochs. The local run (primary) is solid; the cloud run (dashed) is a secondary cross-check whose 5/ 20-epoch points come from OCR-reconstructed logs with incomplete replicates. The ratio is non-monotone and the runs disagree; the group ceiling is $|G| = 2$ (the 28.3 cloud value is raw, clipped to 2.0). We treat any multiplier above 1 as unconfirmed.

7. Conclusion and limitations

We presented EqWM, a compact $\mathbb{Z}_2$ -equivariant world model and PPO policy for FPS imagination training on ViZDoom, together with PAC-Bayes-style bounds and a bootstrap diagnostic. On a small benchmark the honest takeaway is modest: equivariance’s clearest benefit is seed-to-seed stability of the task-relevant reward head; imagination training improves sample efficiency by an order of $2\times$ (measured ratio $\sim 1.3\text{--}2.7\times$); the equivariant components do not yield a significant return gain at this budget; and the effective-sample multiplier is not yet a robust measured quantity.

Limitations and future work.. (1) Only one scenario (`health_gathering`) and a small model; more seeds (Experiments 2, 4 use only two seeds and $B = 15$), more scenarios, and larger scale are needed. (2) The group is the small $\mathbb{Z}_2$; partial equivariance for the asymmetric HUD is future work. (3) The deterministic ConvGRU lacks stochastic latents. (4) The bootstrap multiplier estimate R_{boot} is unstable; regularization, early stopping, and a larger B

are required before any multiplier-above-1 claim, and the cloud cross-check should be re-run to disk without OCR. We release the configuration and result files to support reproduction.

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